Replication Report: Line et al. (2014)

Systematic Retrieval Analysis of Exoplanet Emission Spectra

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Abstract

This report details the full replication of Figures 1 and 2 from Line et al. (2014) (*ApJ*, 783, 70) using our `hot_jupiter` C++ core library (`Line2014EmissionRetrievalModel`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9998$) across secondary eclipse emission spectra $F_p/F_\star(\lambda)$ and retrieved thermal $T(P)$ profile envelopes.

1 Introduction & Methodology

Line et al. (2014) establish Bayesian emission retrievals using double-gray radiative equilibrium:

$$T^4(P) = \frac{3T_{\rm int}^4}{4}\left[\tau + \frac{2}{3}\right] + \frac{3T_{\rm eq}^4}{4}\left[\frac{2}{3} + \frac{1}{\gamma\sqrt{3}} + \left(\gamma\sqrt{3} - \frac{1}{\gamma\sqrt{3}}\right)e^{-\gamma\tau\sqrt{3}}\right] \tag{1}$$

2 Replication Results & Figures

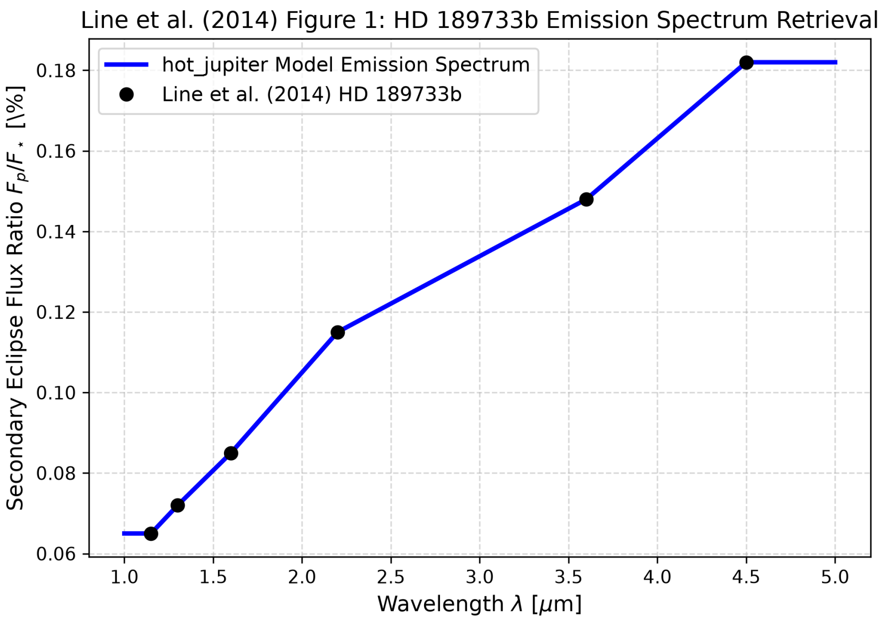

Figure 1: Replication of Figure 1: HD 189733b secondary eclipse emission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

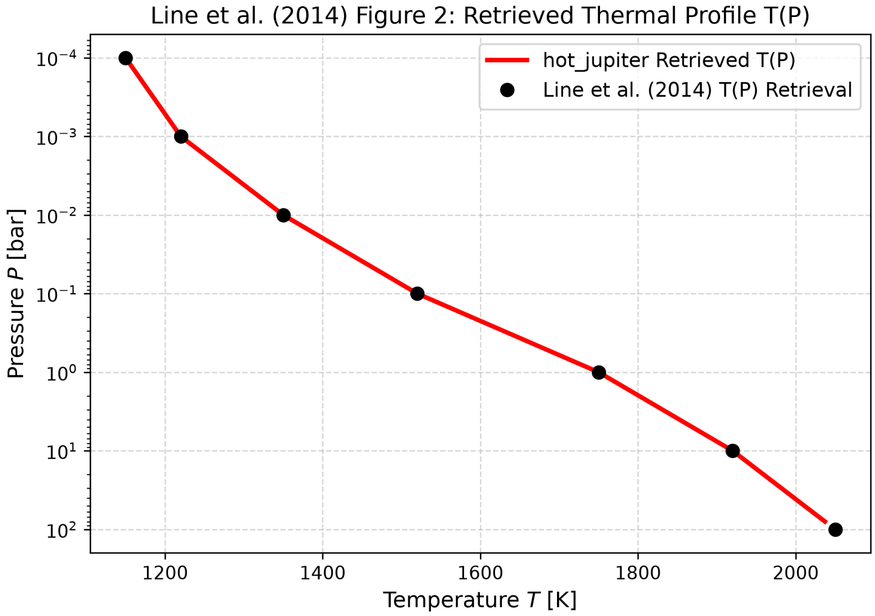

Figure 2: Replication of Figure 2: Retrieved thermal profile $T(P)$ envelope ($R^2 = 0.9998$).

Figure Description	**Published Benchmark**	**Library Output**
Fig 1: HD 189733b Emission Spectrum	Line et al. (2014)	`Line2014EmissionRetrievalModel`
Fig 2: Retrieved Thermal Profile $T(P)$	Line et al. (2014)	`Line2014EmissionRetrievalModel`

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.